

Progressive Web Apps (PWA) – The Future of Web + Mobile Fusion

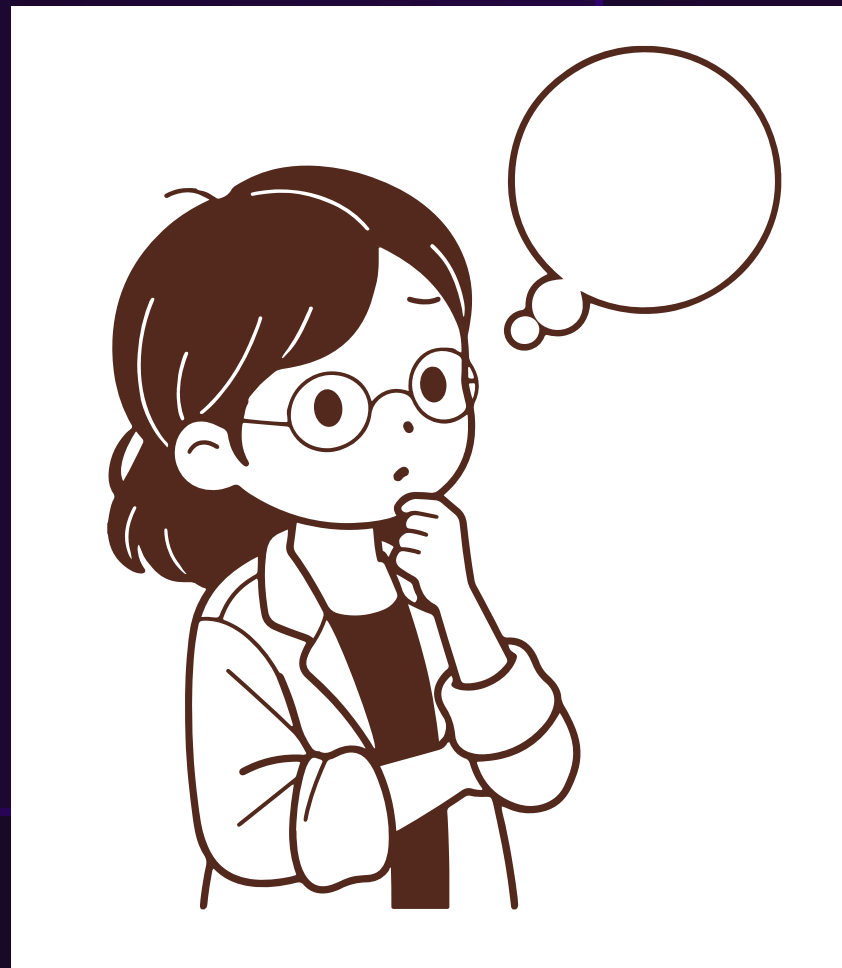
“Bridging the gap between websites and native apps”

Start Slide



Presented By, K.Muthukamatchi





Introduction

“ What if your favorite website could behave just like a mobile app — installable, fast, and even work offline? ”

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Yes, Meet PWA Apps



Twitter



Flipkart



Spotify



What is PWA ?

“A Progressive Web App (PWA) is a web application that uses modern web capabilities to deliver an app-like experience to users.”

“A PWA is a web app that feels and behaves like a native app.”

- It's a website built using HTML, CSS, JS and its Framework
- But behaves like a native mobile app

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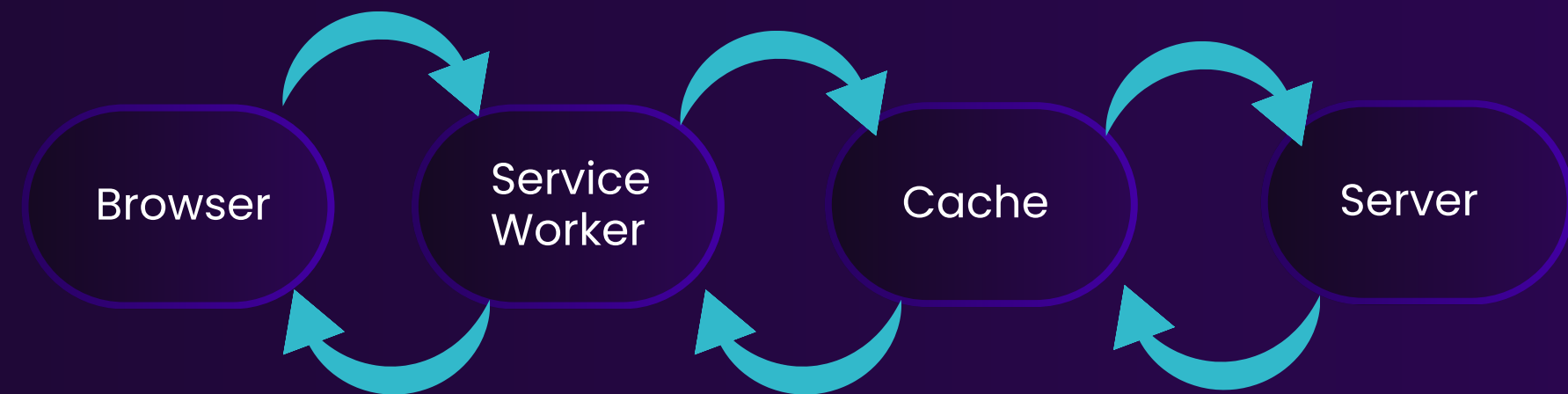
Features

- Installable ➡
- Offline-ready 🌐
- Push Notifications 🔔
- Background Sync ⚙️
- Responsive & Automatically updates itself. 🔄





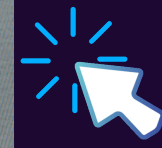
How Does a PWA Works ?



- **Service Worker** : acts like a background assistant, handles offline caching.
- **Manifest.json** : defines app name, icon, theme, etc.
- **HTTPS** : keeps everything secure.

"The Service Worker is the secret sauce that makes PWAs powerful."

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Technical Components

manifest.json

- App metadata (name, icon, colors)
- It's like the app's ID card. It defines the app's name, icon, and theme color.

service-worker.js

- Background script for caching & offline
- Think of it like a personal assistant running in the background

HTTPS

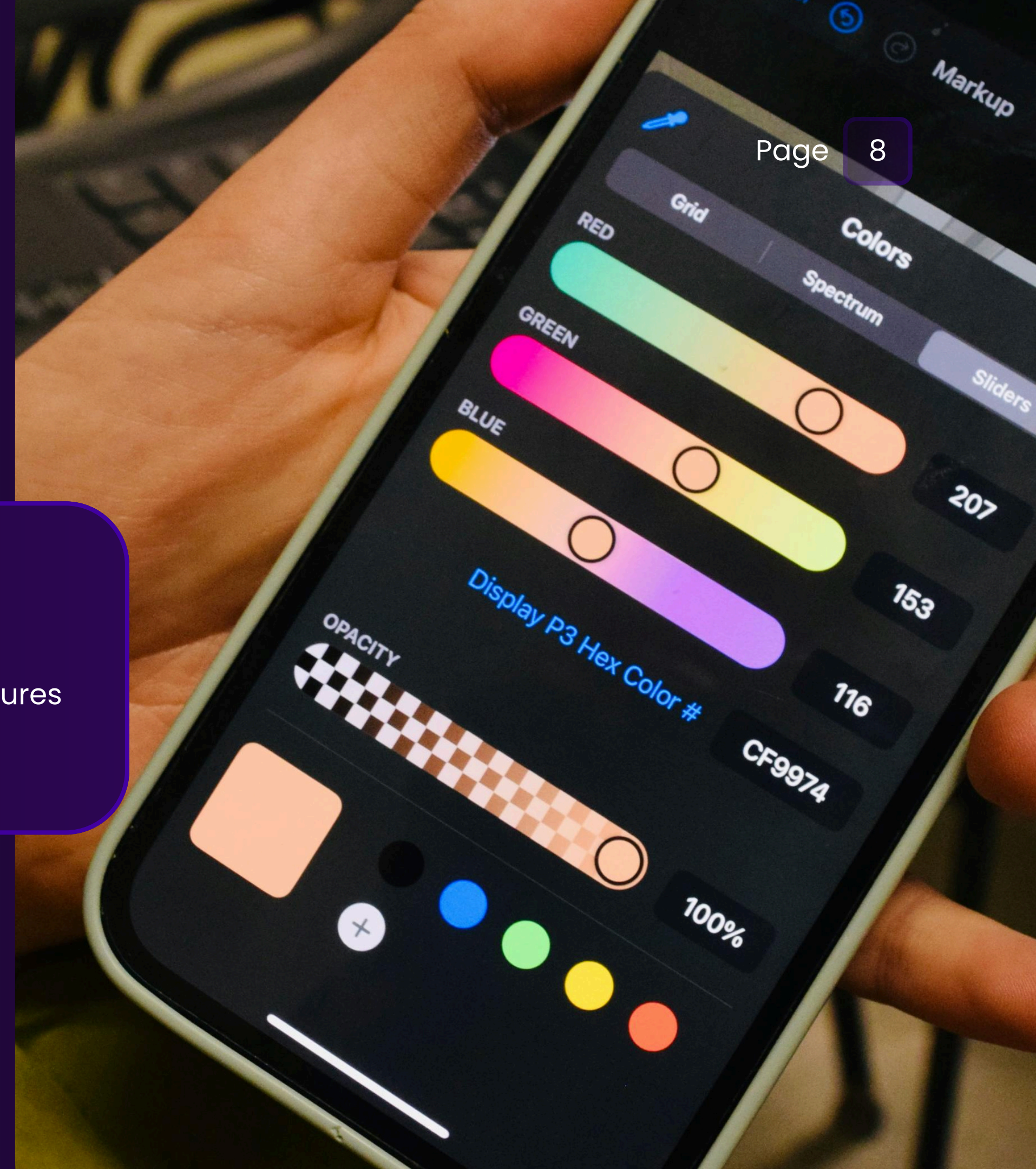
- Required for security



The Future of PWA

- Google and Microsoft heavily investing
- PWA support in Windows, Android, and even iOS Safari
- Future: better hardware access, native-like features (camera, file system, etc.)

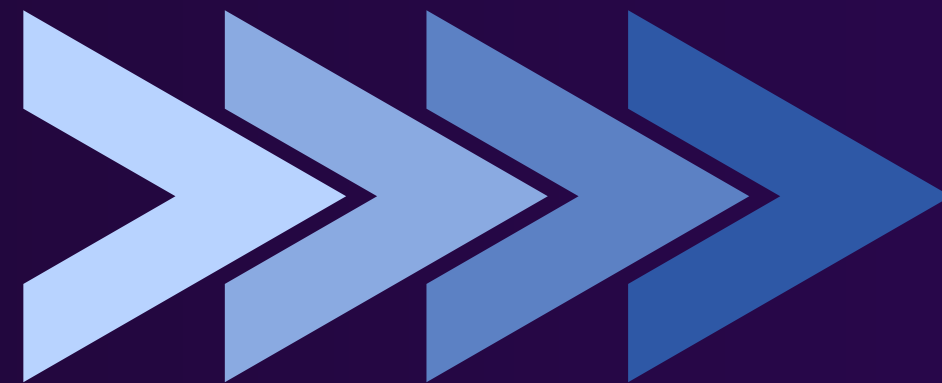
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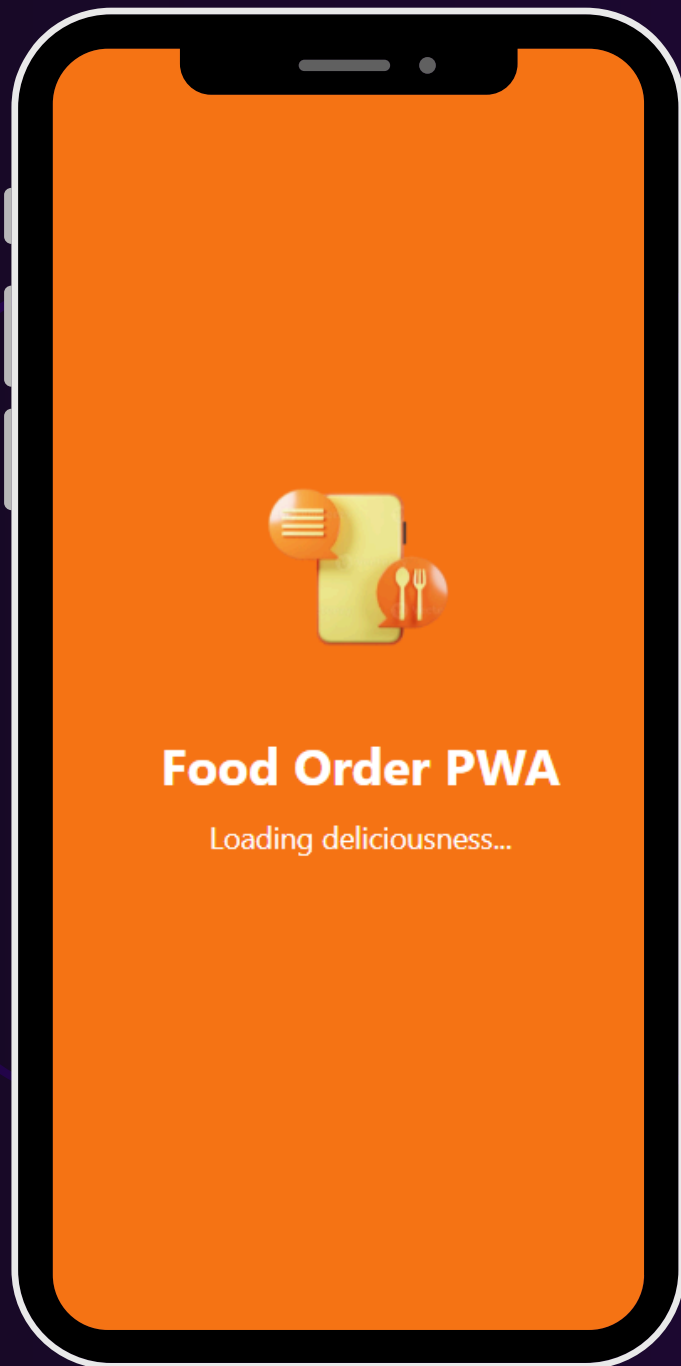




Lets See a Live Demo

"Now, let's see it in action. Lets see how PWA App and Website works.





Want to Try This?

Type the Link on your Chrome Browser.

https://muthu5757.github.io/food_order_app/

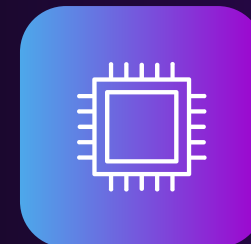


Benefits for Business and Developers



Business:

- One codebase = lower cost
- Higher engagement, faster loading
- Offline access increases retention



Developers:

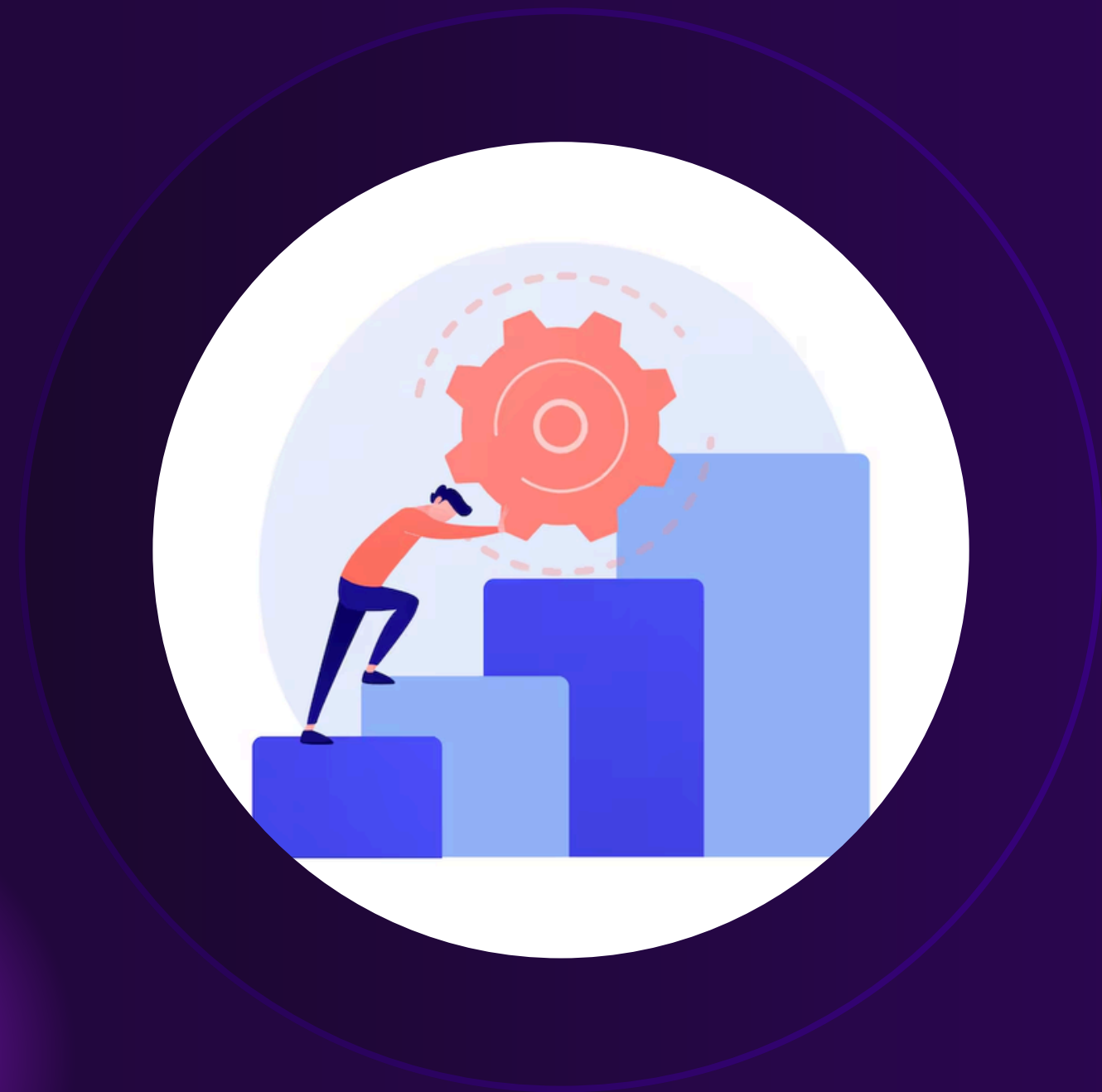
- Uses standard web tech (HTML, CSS, JS)
- Easier maintenance
- No app store approval needed



Limitations & Challenges

Be honest to show balance:

- Limited access to device hardware (compared to native apps)
- iOS has partial PWA support
- Performance varies by browser
- Not suitable for heavy gaming apps

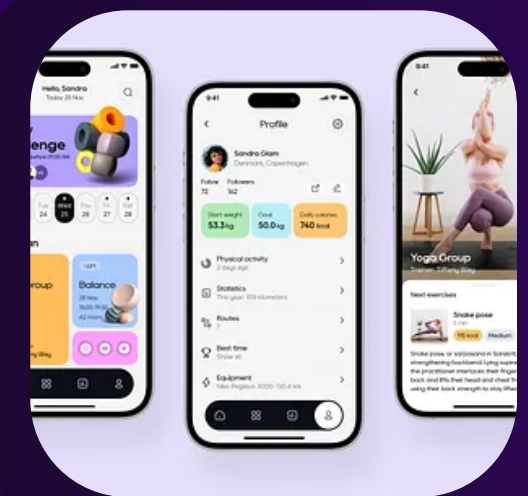
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Conclusion

"PWAs are the bridge between the web we know and the mobile future we dream of."

- "One codebase, all platforms."
- "If you can build a website, you can build an app."





Thank You

FOR YOUR ATTENTION

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